Permutations & Combinations.

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↳ First of all you must know about factorial.

↳ Factorial:- Factorial of a number nothing but product of the given number to '1' by decreasing consecutively (or) successively by '1'.

Factorial represented as '!' (or) 'L'

Ex:- $n! = n(n-1)(n-2)(n-3)\ldots\ldots\times 1$

$6! = 6\times5\times4\times3\times2\times1 = 720$

Note!- $0! = 1$

↳ Permutations:- Arrangement (or) re-arrangement of objects in given places. Here mainly order of objects in arrangements are different. No two arrangements are same in permutations.

Here total two types of arrangements we generally see, they are without repetation & with Repetation.

Ex:- Arrangement of A, B & C in 3 places without repetation [default case].

A, B, C B, C, A
A, C, B C, A, B
B, A, C C, B, A

} Total 6 ways we can arrange w/o repetation.

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(or) A, B, C

$$\underset{A/B/C}{\underline{3}} \times \underset{\begin{matrix}A/B\ (\text{if } C)\\ B/C\ (\text{if } A)\\ C/A\ (\text{if } B)\end{matrix}}{\underline{2}} \times \underset{\begin{matrix}A\ (\text{if } B \& C)\\ B\ (\text{if } A \& C)\\ C\ (\text{if } A \& B)\end{matrix}}{\underline{1}} = 6 \text{ ways}$$

$= 6 \text{ ways} = 3 \times 2 \times 1 = 3! = {}^3P_3$

Permutation (w/o Repetation or default) :- 'n' objects are to be arranged in 'r' places ($r \le n$) without repetition in ${}^nP_r$ ways.

$$\boxed{{}^nP_r = \frac{n!}{(n-r)!}}$$

Ex:- How many ways we can give ranks to 3 persons in running race?

$\underline{3}\ \underline{2}\ \underline{1} = 6$ ways

Permutation with Repetation :- If n objects are to be arranged in given places w/o conditions, in such a way that any one of all objects can be placed in every place of given places.

Ex:- Arrange A, B, C in 3 places with repetition :-

$$\underset{A/B/C}{\underline{3}}\ \ \underset{A/B/C}{\underline{3}}\ \ \underset{A/B/C}{\underline{3}} = 3 \times 3 \times 3 = 3^3 = 27$$

here objects are '3' & places are also '3'.

Ex:- How many ways 20 students of a class can get 1st rank in 3 subjects?

Ans:- Here there is no way of w/o repetition, only repetation, we have to consider as default.